

Dataset Analysis for Decision-Making

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sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and provide a clear, concise answer to any specific analytical question, I will first examine the structure and key characteristics of the dataset. This includes identifying column names, data types, missing values, and general patterns in the data.

Below is the Python code to load and inspect the dataset:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
14
15 # Check for missing values
16 print("\nMissing Values per Column:")
17 print(data.isnull().sum())
18
19 # Summary statistics for numerical columns
20 print("\nSummary Statistics for Numerical Columns:")
21 print(data.describe())
```

Key Characteristics of the Dataset:

- Column Names** : The `data.info()` output will list all column names and their data types.
- Data Types** : Identifying whether columns are categorical (e.g., strings) or numerical (e.g., integers, floats).
- Missing Values** : Checking for null or missing entries in each column.
- Numerical Summary** : Using `data.describe()` , we can identify min, max, mean, median, and other statistical properties of numerical columns.

Once you provide a specific analytical question, I will use this dataset to extract relevant information and provide a precise answer with accompanying Python code.

Please provide your analytical question now.

What factors are associated with some economic activities receiving higher investment values than others?

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To address the question of what factors are associated with some economic activities receiving higher investment values than others, we need to analyze the dataset for patterns and relationships. This involves examining how investment values (likely a numerical column) correlate with other variables such as economic activity type, geographic location, contractor



entity, a numerical column correlates with other variables such as economic activity type, geographic location, contractor details, or other relevant features.

Here's how we can approach this:

Steps to Analyze the Data:

1. Identify Relevant Columns :

- Locate the column that represents "investment values" (e.g., contract amounts or financial allocations).
- Identify columns that describe "economic activities" (e.g., category, sector, or description of the activity).

2. Group and Aggregate Data :

- Group the data by economic activity and calculate summary statistics (e.g., mean, median, total) for investment values.

3. Explore Relationships :

- Determine if certain economic activities consistently receive higher investments.
- Investigate whether additional factors (e.g., geographic location, year, or contractor type) influence investment levels.

4. Visualize Insights :

- Use bar charts or box plots to visualize differences in investment values across economic activities.

Python Code to Analyze the Data:

Below is the Python code to perform the analysis:

```
python
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the dataset
5 file_path = 'sicor_municipal_contracts.csv'
6 data = pd.read_csv(file_path)
7
8 # Identify relevant columns
9 # Assuming 'investment_value' is the column for investment amounts and 'economic_activity' is the
10 investment_column = 'investment_value' # Replace with actual column name
11 activity_column = 'economic_activity' # Replace with actual column name
12
13 # Check if the assumed columns exist
14 if investment_column not in data.columns or activity_column not in data.columns:
15     raise ValueError(f"Columns '{investment_column}' or '{activity_column}' not found in the data")
16
17 # Group data by economic activity and calculate total and average investment values
18 activity_summary = data.groupby(activity_column)[investment_column].agg(['sum', 'mean', 'count'])
19 activity_summary.columns = [activity_column, 'total_investment', 'average_investment', 'count']
20
21 # Sort activities by total investment in descending order
22 activity_summary = activity_summary.sort_values(by='total_investment', ascending=False)
23
24 # Display the top economic activities by total investment
25 print("\nTop Economic Activities by Total Investment:")
26 print(activity_summary.head())
27
28 # Visualize the top economic activities by total investment
29 plt.figure(figsize=(10, 6))
30 top_activities = activity_summary.head(10) # Top 10 activities
31 plt.barh(top_activities[activity_column], top_activities['total_investment'], color='skyblue')
32 plt.xlabel('Total Investment')
33 plt.ylabel('Economic Activity')
34 plt.title('Top Economic Activities by Total Investment')
35 plt.gca().invert_yaxis() # Invert y-axis for better readability
36 plt.show()
37
38 # Explore additional factors (e.g., geographic location)
39 # Assuming there's a 'location' column
40 if 'location' in data.columns:
41     location_summary = data.groupby(['location', activity_column])[investment_column].sum().reset_index()
42     location_summary = location_summary.sort_values(by=investment_column, ascending=False)
43     print("\nInvestment by Location and Economic Activity:")
44     print(location_summary.head())
```

Explanation of the Code:



1. Column Identification :

- The code assumes `investment_value` and `economic_activity` as the relevant columns. You may need to replace these with the actual column names from your dataset.

2. Grouping and Aggregation :

- The dataset is grouped by `economic_activity`, and summary statistics (`sum` , `mean` , `count`) are calculated for `investment_value` .

3. Sorting and Visualization :

- The top economic activities are identified based on total investment and visualized using a horizontal bar chart.

4. Additional Factors :

- If geographic location or other factors are available, the code explores their relationship with investment values.

Hypothetical Insights:

- Certain economic activities (e.g., infrastructure, healthcare) might receive significantly higher investments due to their importance or scale.
- Geographic location could play a role, with urban areas attracting more investment than rural areas.
- The number of contracts or projects in each activity type might also influence total investment.

Next Steps:

- Run the code and verify the results.
- Provide the actual column names for `investment_value` and `economic_activity` if they differ from the assumptions.
- Share any additional factors you'd like to explore (e.g., time trends, contractor details).

Let me know the correct column names or any refinements needed!

